STEAMROOT · PROBLEM STATEMENT

SteamRoot problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-25	Amish Chadha	Users-and-needs and stakeholder tables added (kit 1.9.1 upgrade, STANDARDS section 22); user requirements in STR-URS-001
0.5	2026-10-02	Amish Chadha	Partners and authority answered by Amish's 2026-10-02 decisions
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on first users, first jurisdiction and budget; regulatory ruling marked as an open prerequisite
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (users, context, constraints, out of scope, prior work, safety)
0.1	2026-09-24	Amish Chadha	Initial scaffold

SteamRoot problem statement

Steaming soil kills weed seeds and soil-borne pathogens without herbicides or fumigants, but the equipment that does it is sized, priced and fueled for large greenhouse operations, so small organic growers who need it most rarely have access to it.

The problem

Organic and low-input growers fight weeds and soil-borne disease (damping-off, Fusarium, Pythium, Rhizoctonia, root-knot nematodes) with hand weeding, cultivation, rotation and, where allowed, fumigants. Hand weeding is often the largest labor cost on a market garden. Chemical soil fumigants are restricted or banned for organic growers, and methyl bromide has been phased out for most uses under the Montreal Protocol.

Heating soil to about 70 °C (158 °F) for 20 to 30 minutes kills most weed seeds and pathogens while sparing some of the heat-tolerant beneficial organisms. Steam is the traditional way to deliver that heat because condensing steam releases a large amount of energy (about 2.26 MJ per kilogram) right where it condenses. The barriers are equipment and energy:

- Commercial soil steamers are built around diesel, oil or gas boilers sized for greenhouse ranges and nurseries. They are costly to buy and to fuel.
- Many small growers have surplus woody biomass (prunings, hedgerow wood, pallets) but no safe way to turn it into steam.
- Improvised rigs built from pressure cookers, water heaters or sealed tanks are dangerous. A sealed vessel with a fire under it can fail violently.
- Energy per square meter is high, so an inefficient boiler makes the method too slow and too fuel-hungry to be practical.

Users and context

User	Need	Context
Small organic farm or market garden	Clear weed seed bank and disease from high-value beds without herbicides	Outdoor beds and polytunnels, 0.1 to 2 ha, beds about 0.75 to 1.2 m wide
Nursery or seedling grower	Pasteurize potting media and propagation beds	Sheds and greenhouses; generator must stay outside
Community garden or farm cooperative	A shared, towable machine that several growers can use	Shared equipment, varied operator experience
Extension educator or researcher	An open, documented reference design to demonstrate and study	Field days, trials, university farms
Open hardware community	A buildable, inspectable low-pressure steam design	Farm workshops, makerspaces

Table 1. Users and needs (kit 1.8 form; the user requirements drawn from it are in STR-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Small organic farm or market garden	Clear weed seeds and disease from high-value beds without herbicides; use the farm's own wood	Steaming beds 0.75 to 1.2 m wide before a high-value crop, hood by hood, through a working day	Hand weeding is often the largest labour cost; commercial steamers are diesel-fired and costly
Nursery or seedling grower	Pasteurize propagation beds and potting media	Steam piped into a polytunnel or shed from a generator parked outside	The fire and its carbon monoxide must stay outdoors
Community garden or farm cooperative	Share one towable machine between growers	Towed between sites by car or utility vehicle; varied operators	Operator skill varies; towed mass and width limits
Operator (any of the above)	Run it safely alone with a helper nearby	Lighting, setting the feed and dampers, moving the hose and hoods	Steam burns, hot surfaces, heavy hoods, refilling water
Extension educator or researcher	An open, documented reference design	Field days, trials, university farms	Proprietary equipment cannot be inspected or taught from
Open hardware community	A buildable, inspectable low-pressure steam design	Farm workshops and makerspaces	Improvised sealed rigs are dangerous

Stakeholders

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Texas Department of Licensing and Regulation, boiler program	Rules whether an open-vented, fired monotube coil is a regulated boiler (Texas Health and Safety Code chapter 755)	A written ruling before any build (decided by Amish 2026-10-02); the design is reviewed against it
Texas A&M AgriLife Extension, horticulture contacts	Route to the first co-design partners (decided by Amish 2026-10-02)	Introductions to one market garden and one nursery
First co-design partners (one market garden, one nursery near Irving)	Use the machine and judge the targets	Review the user and product requirements; host TRL 4 soil tests when Amish approves TRL 4
County fire marshal or commissioners court	Burn bans on outdoor burning in dry periods (Texas Local Government Code 352.081)	Check before each use (STR-REG-001)
Organic certifier	Whether steaming fits the grower's organic system plan (7 CFR 205.206 lists heat as a weed control)	The grower confirms with their certifier
Amish Chadha	Owner of scope, budget, naming and safety trade-offs	Decisions in STR-DEC-001

Constraints

- Garage-buildable prototype within $2,200 USD (raised from $1,500 to $1,800 and then to $2,200 by Amish on 2026-09-25 to allow about 15 % contingency over the priced BOM), using stock steel, stainless tube and off-the-shelf safety parts.
- Near-atmospheric operation: open-vented, never sealed, with a normal working pressure well below the thresholds that bring a boiler under pressure vessel and boiler codes. Thresholds, exemptions and registration rules differ by country, state and province and must be confirmed with the local authority before any build.
- Fired with locally available dry woody biomass; no diesel or propane burner.
- Towable by a small tractor, utility vehicle or car with a hitch; 500 kg or less as towed, with the feed tank and seal pot drained and filled on site.
- Operable by one trained adult, with a second person nearby during steaming.
- Outdoor operation only for the fire. Steam may be piped into a greenhouse or polytunnel; the firebox and chimney may not.

> **Safety:** This project handles open fire, flue gas containing carbon monoxide, and steam that causes severe burns. It is a concept, not a buildable or approved design. Steam generators are regulated equipment in many jurisdictions even at low pressure.

Out of scope

- Any pressurized boiler or sealed vessel. SteamRoot will not operate as a pressure boiler under any configuration.
- Deep steaming beyond 15 cm and field-scale broadacre treatment.
- Negative-pressure (buried pipe) steaming systems.
- Automatic fire control, remote operation or unattended running.
- Chemical additives mixed with steam.

Prior work

Steam soil disinfestation is established practice. Greenhouse growers have steamed soil since the late nineteenth century, and it remains common in greenhouse horticulture and nurseries. The main methods are:

- **Sheet steaming:** steam is released under a heavy sheet laid on the soil. Simple, but slow and hard to control at depth.
- **Hood steaming:** a rigid open-bottom hood is pressed onto the bed and moved in steps. Better contact and control than sheets. SteamRoot uses this method.
- **Negative-pressure steaming:** steam is drawn down through the soil by suction through buried perforated pipes. Reaches deeper, but needs a permanent installation.
- **Band steaming:** research machines steam only a narrow band where the crop row will be, cutting energy per hectare.

Soil solarization (clear plastic in hot weather) and anaerobic soil disinfestation are lower-energy alternatives, but they depend on climate and take weeks. Steam works in hours and in any season. A 2014 UNEP review of the methyl bromide phase-out called steam "probably the best technical alternative to MB in protected agriculture", but noted that deep in-ground steaming needs long boiler use, labor and fuel that can make it uneconomic (UNEP

Ozone Secretariat, *Phasing-out Methyl Bromide in Developing Countries*, 2014). Energy per square meter is therefore the main limit on adoption, which is why SteamRoot focuses on a biomass fire and a flue-gas economizer. The other prior-work statements above are general practice and are not yet individually cited.

Decisions and open questions

- First users: one market garden and one nursery. Decided by Amish, 2026-09-25 (STR-DDR-001). Decided by Amish, 2026-10-02: one market garden and one nursery within driving distance of Irving that already steam, solarize or chemically treat soil, found through Texas A&M AgriLife Extension's horticulture contacts. None is named or agreed yet.
- First jurisdiction for the regulatory check: Amish's home jurisdiction. Decided by Amish, 2026-09-25. Decided by Amish, 2026-10-02: Texas; the ruling is asked of the Texas Department of Licensing and Regulation's boiler program.
- **Open prerequisite:** a written ruling from the local boiler authority on whether an open-vented, fired monotube coil is exempt from boiler and pressure vessel rules. Amish approved getting it before TRL 3 detail work. It has not been obtained, so TRL 3 was limited to paper calculations and preliminary drawings.